

Please answer the ff. questions:

• Part A: Variable Identification Table

Variable name	Data Type	Variable Type	Stored In
name	String	Instance	Heap (inside Student)
grade	int	Instance	Heap (inside Student)
n	String	Local / Parameter	Stack
g	int	Local / Parameter	Stack
args	String[]	Local / Parameter	Stack
s1	Student	Local / Object Reference	Stack
s2	Student	Local / Object Reference	Stack
s3	Student	Local / Object Reference	Stack

• Part B: Explanation. Please see example below:

Statement	What happens?	Memory Impact
Student s1 = new Student("John", 85);	A new student object is made with the name John and grade of 85.	Stack: Allocates space for s1 variable. Heap: Stack points to this new heap.
Student s2 = new Student("Maria", 90);	A second student is made with the name Maria and grade of 90.	Stack: Allocates space for s2 variable. Heap: Stack points to this new heap.
Student s3 = s2;	The information in s2 is copied over to s3.	Stack: Allocates space, but copies address. Heap: No new object is created.
s1.grade = 95;	The grade property of the object referenced by s1 is changed to 95.	Stack: References don't change. Heap: Data is overwritten with 95.
s3.name = "Ana";	The name of the object referenced by s3 is updated to Ana.	Stack: References don't change. Heap: Data is overwritten with Ana.

• Part C - Output

John 95

Ana 90

Ana 90